

1.3: Equations and Graphs of Polynomial Functions

Functions may be expressed in many ways. For example a quadratic equation may be expressed either in standard form, factored form or vertex form. The same is true for polynomial functions.

ex: $f(x) = x^3 + 2x^2 - 5x - 6$ may also be written in factored form as $f(x) = (x + 1)(x - 2)(x + 3)$

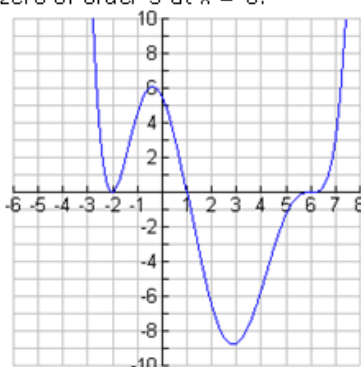
Factored form is very useful when graphing polynomial functions since the zeros can easily be determined from the factors (it is no longer necessary to create a table of values).

Recall: The zeros of a polynomial function $y = f(x)$ correspond to the x -intercepts of the graph and to the roots of the corresponding equation $f(x) = 0$.

New: Order (of a zero): The number of times a factor $(x - t)$ is repeated.

Example: $f(x) = (x - 1)(x + 2)^2(x - 6)^3$, has a zero of order 1 at $x = 1$, a zero of order 2 at $x = -2$ and a zero of order 3 at $x = 6$.

$x = -2$ is a zero of order 2; the sign of the function does not change at this x -intercept, it remains positive.



$x = 1$ is a zero of order 1; the sign of the function changes at this x -intercept from positive to negative.
 $x = 6$ is a zero of order 3; the sign of the function changes at this x -intercept from negative to positive and there is a slight wave as it does so.

NOTES:

- A function is **positive** when it lies above the x -axis and **negative** when it lies below the x -axis.
- The graph changes **sign** at zeros that are of odd order but **not** at zeros that are of even order.

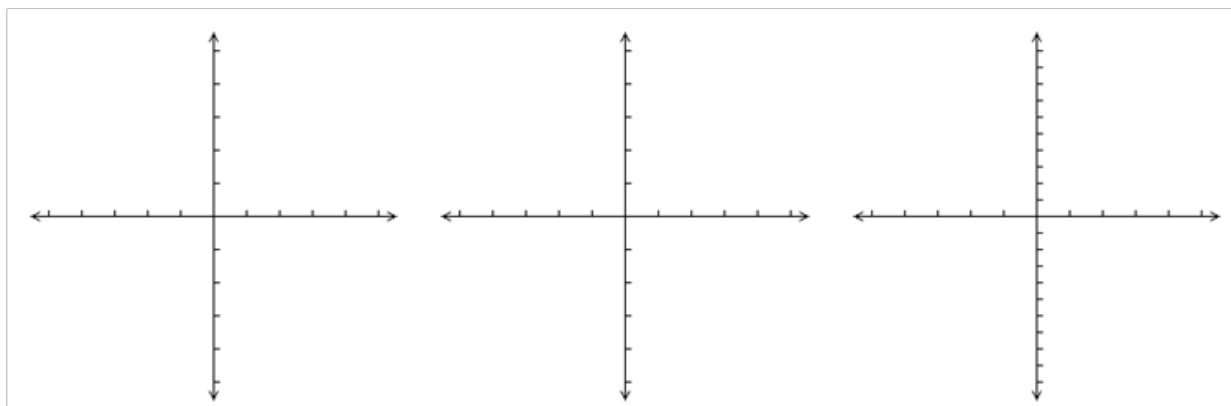
Example 1:

Sketch a graph for each polynomial function. Make sure to include all intercepts.

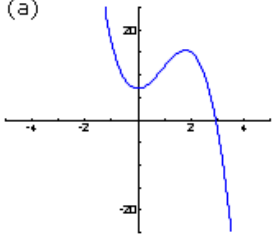
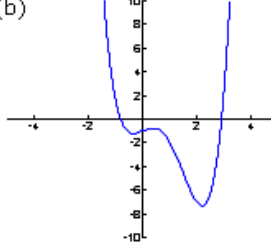
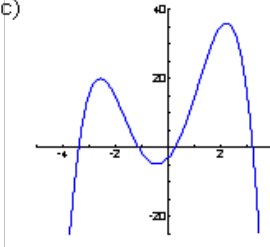
a) $y = -2(x + 1)^3(x - 1)^2$

b) $y = (x + 1)(x - 1)^2(x + 3)$

c) $y = -(x^2 - 2x - 3)(2x^2 - 7x + 3)$



Example 2: Use the graph of each polynomial to identify the function as **cubic** (degree = 3) or **quartic** (degree = 4). State the sign of the leading coefficient, describe the end behaviour, and state the intervals where the function is positive and negative.

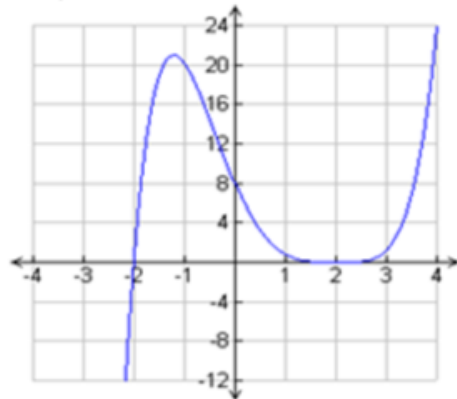
	(a) 	(b) 	(c) 
Type			
Sign of leading coefficient			
End Behaviour			
Positive Intervals			
Negative Intervals			

Example 3: Determine the equation of the following functions in factored form .

a) A cubic function with zeros -2, 1, 4 and a y-intercept of 24.

b) A quartic function with zeros -2, 0 (order 2), 1 and passes through the point (-3, -12).

c) A quintic function as illustrated below.



Homework: p. 39 # 1, 2, 6, 9, 11a
#6c wrong answer. Should be $a = -3$